

|                   |                               |
|-------------------|-------------------------------|
| Exp.<br><b>03</b> | <b>Study of Pipe Friction</b> |
|-------------------|-------------------------------|

**Formulas:**

| For Turbulent Flow           |                                                    |
|------------------------------|----------------------------------------------------|
| $h_{f, \text{Experimental}}$ | $h_{f(\text{exp})} = 12.6h$                        |
| Reynolds Number, Re          | $\text{Re} = (\rho V D) / \mu$                     |
| Friction factor, f           | $f = 0.316 \text{Re}^{-0.25}$                      |
| $h_{f, \text{Theoretical}}$  | $h_{f(\text{th})} = f \frac{L_e}{D} \frac{V^2}{2}$ |

| Flow Type      | Expected (n)   |
|----------------|----------------|
| Laminar flow   | $n \approx 1$  |
| Turbulent flow | $n = 1.75 - 2$ |

**Observation**

Diameter of pipe, D =

Length of the pipe, L =

Specific weight of flowing fluid (water),  $\gamma =$ Specific weight of manometer fluid (Hg),  $\gamma_{hg} =$ 

Water temperature, t =

Dynamic viscosity,  $\mu =$ **Experimental Data Sheet**

| No. of Obs | U-tube manometer reading for head loss |                           |                       | Rotameter reading    |
|------------|----------------------------------------|---------------------------|-----------------------|----------------------|
|            | Left Column reading (mm)               | Right column reading (mm) | Net deflection h (mm) | Flow rate, Q (L/min) |
| 1          |                                        |                           |                       |                      |
| 2          |                                        |                           |                       |                      |
| 3          |                                        |                           |                       |                      |
| 4          |                                        |                           |                       |                      |
| 5          |                                        |                           |                       |                      |
| 6          |                                        |                           |                       |                      |

## Calculation Data Sheet

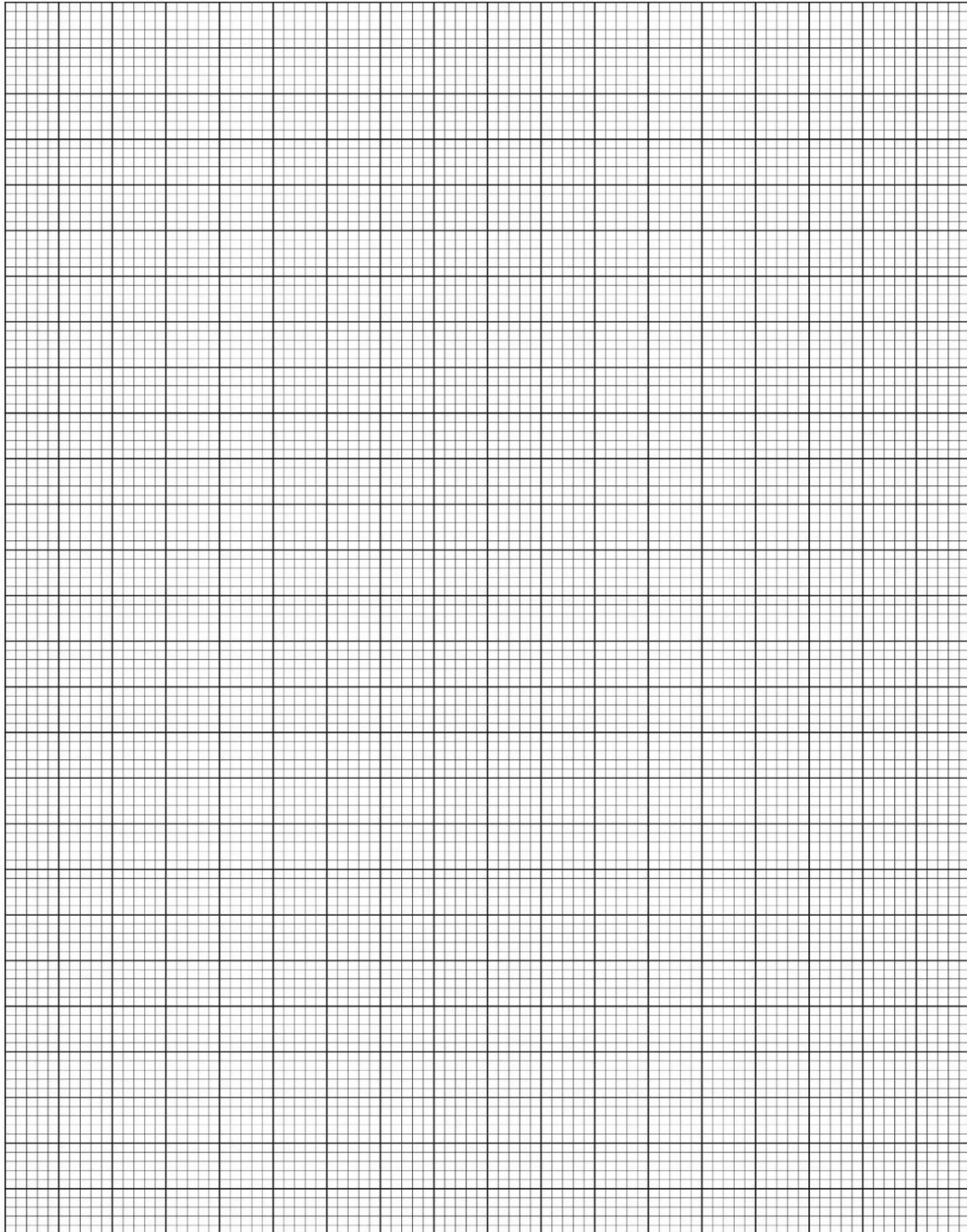
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**Plot:**

- Plot  $\log(h_{f(\text{exp})})$  vs  $\log(Q)$

Along Horizontal Axis:

Along Vertical Axis:



**Sample Calculation:**

Show a sample calculation for one just observation and find the slope of the graph.

**Discussion:**

- Give reasons for discrepancies between the theoretical and measured values of the results.
- Discuss the variation of friction factor with Reynolds number.